5 V Mobile Charger

IIT HYDERABAD

DEPARTMENT OF ELECTRICAL ENGINEERING

EE1010

MATHS FOR ELECTRICAL ENGINEERING

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Block Diagram:

230 V AC → 12 V AC → Rectifier → Capacitor → 7805 → 5 V USB

Objective:

To design and develop a simple AC-to-DC power supply that converts 230 V AC mains into a stable 5 V DC output for powering or charging USB devices using a step-down transformer, bridge rectifier, filter capacitor, and 7805 voltage regulator.

1.Components and their functions

Component	Functions
230 V → 12 V transformer	Steps down and isolates the main voltage.
4 diodes	Form a bridge rectifier and convert AC into pulsating DC.
Capacitor	Smooths the pulsating DC.
7805 Regulator	Regulates the DC to approximately 5 V.
Female USB socket	Provides the 5 V output connection.
Perfboard and wires	Used to assemble the circuit.

2. Overall working:

230 V AC

↓

Transformer (230 V → 12 V)

↓

12 V AC

↓

4-diode bridge rectifier

↓

Pulsating DC

↓

Smoothing capacitor

↓

Smoothed DC

↓

7805 regulator

↓

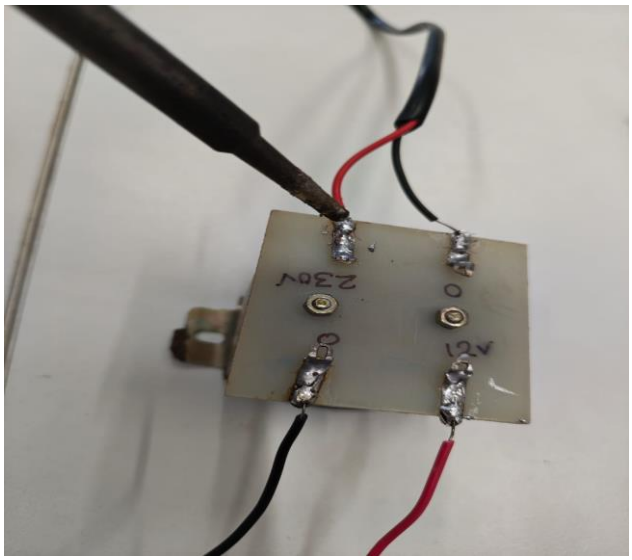
≈ 5 V DC

↓

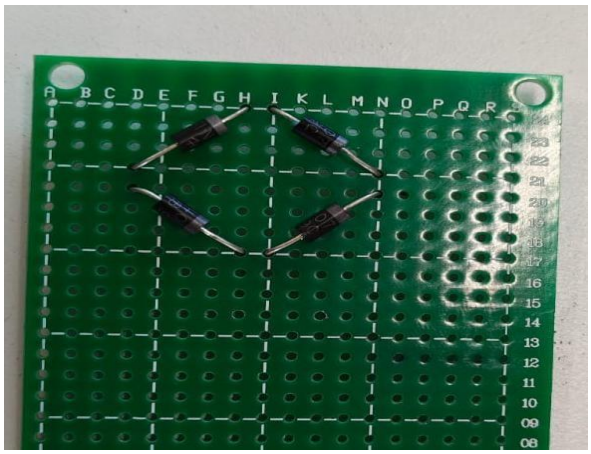
Female USB socket

3. Progress log of circuit:

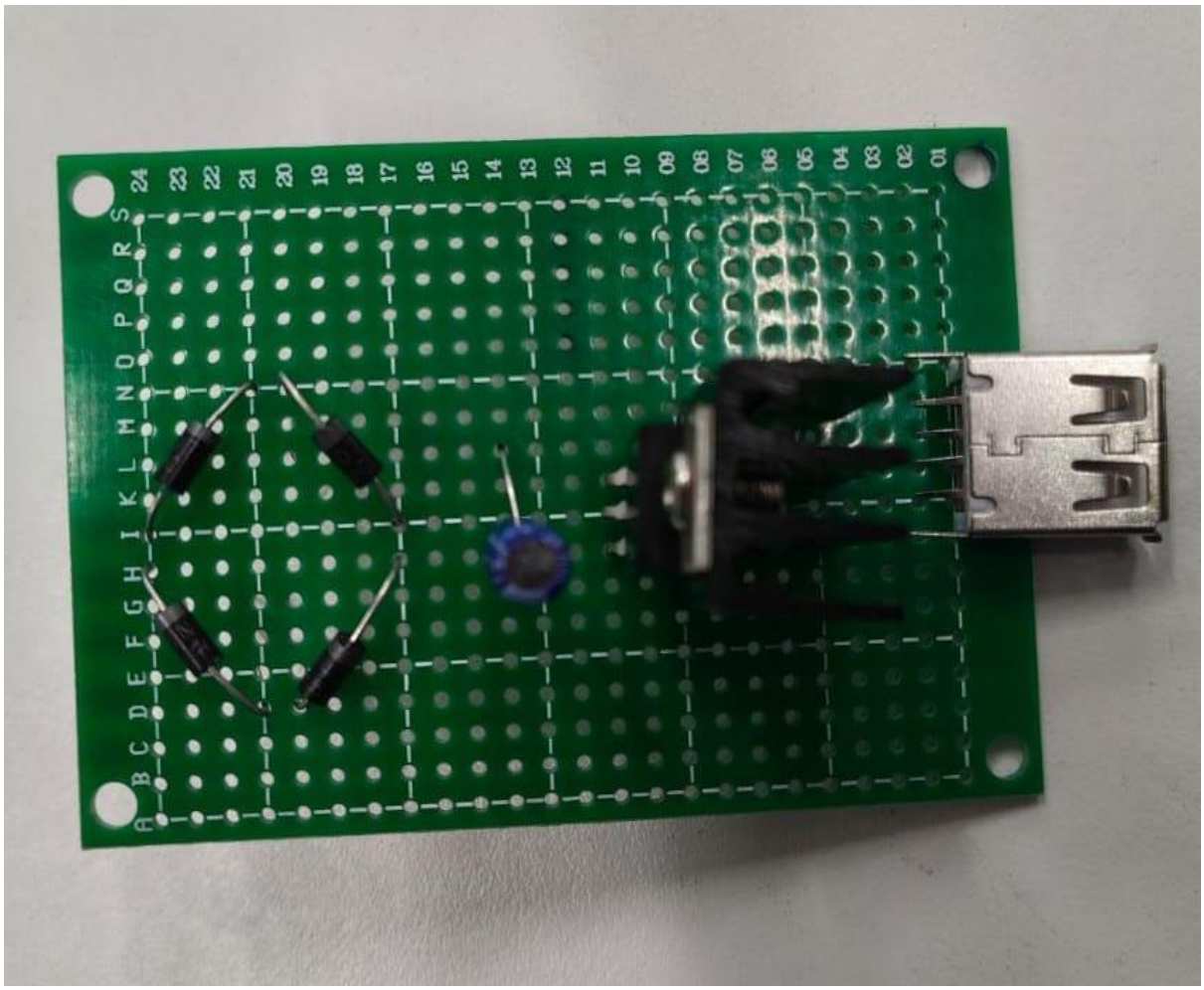
1. At first, we soldered the input cable wires and output wires of the transformer to the transformer's terminal, ensuring a firm and reliable connections as they are not well soldered before.



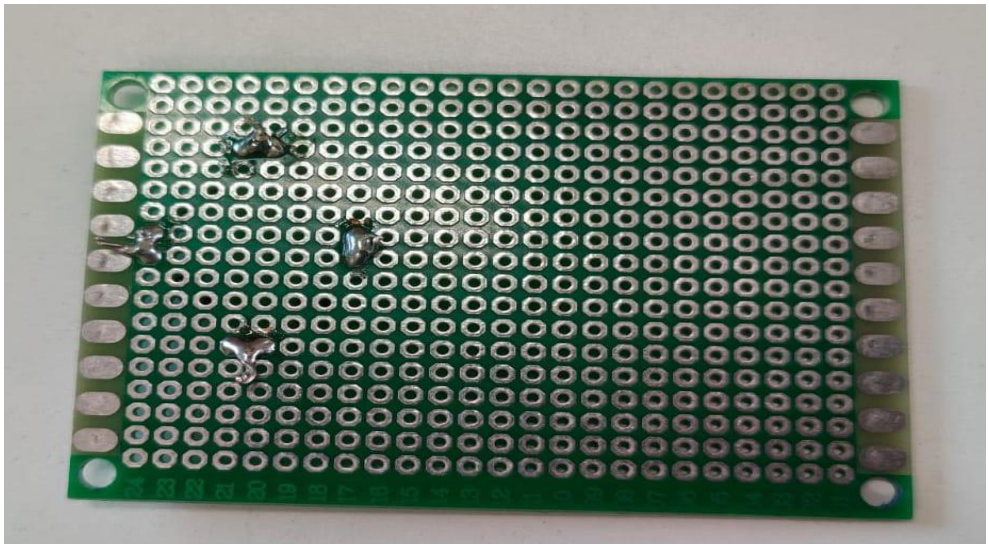
2. The 4 diodes were carefully positioned and oriented in the correct directions to form full wave bridge rectifier, ensuring proper current flow and correct +ve and -ve DC output.



3.The components were temporarily arranged on the perf board to check their spacing, placement & the availability of suitable connection points before soldering.



4. The 4 diodes were soldered together by connecting the appropriate terminals of each diode to form the required bridge rectifier configuration.



5. The capacitor terminals were bent through the perfboard holes and soldered in place to securely mount the capacitor and establish proper electrical connections.

6. The positive terminal of the capacitor was soldered to the junction of the two diode cathodes.

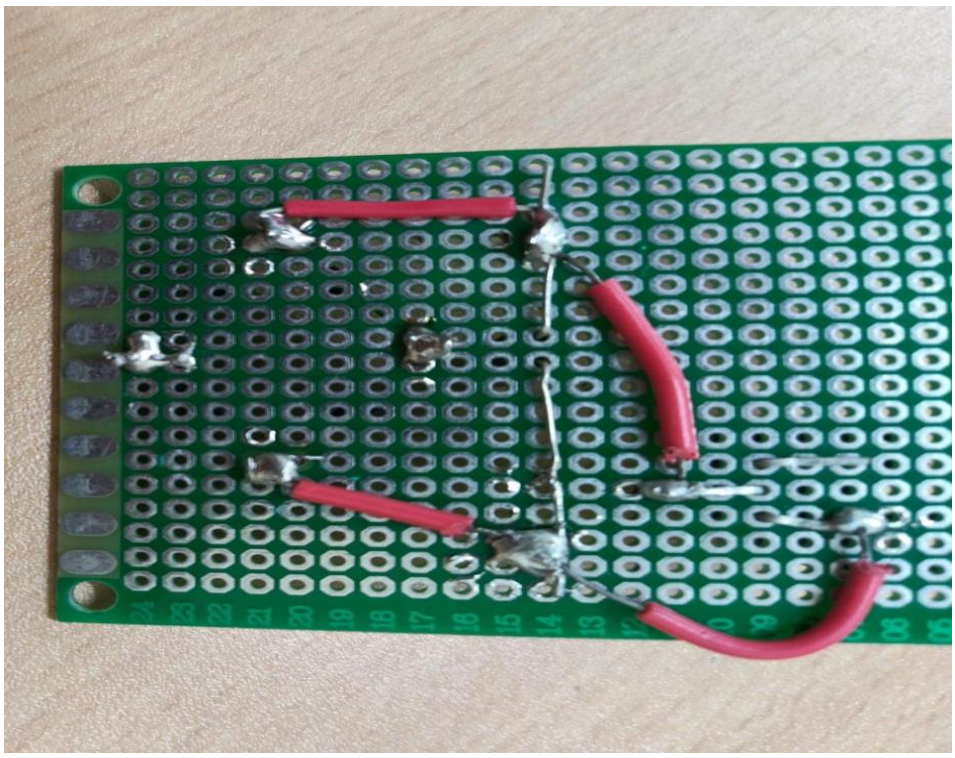
7. The negative terminal of the capacitor was soldered to the junction of the two diode anodes, forming the positive DC output of the bridge rectifier.

8. The two output wires from the transformer were connected and soldered to the corresponding AC input terminals of the diode bridge to provide the 12 V AC supply for rectification.

9. The voltage across the capacitor was measured using a multimeter to verify the rectified and filtered DC output.

10. The 7805-voltage regulator was placed onto the perfboard in a convenient position, allowing the input, ground, and output connections to be clearly routed to the rectifier and USB socket using connecting wires.

11. The positive terminal of the capacitor was connected to the input (IN) pin of the 7805 voltage regulator, while the negative terminal was connected to the ground (GND) pin.



12. The output voltage of the 7805-voltage regulator was measured using a multimeter to verify that it provided a stable 5 V DC output.

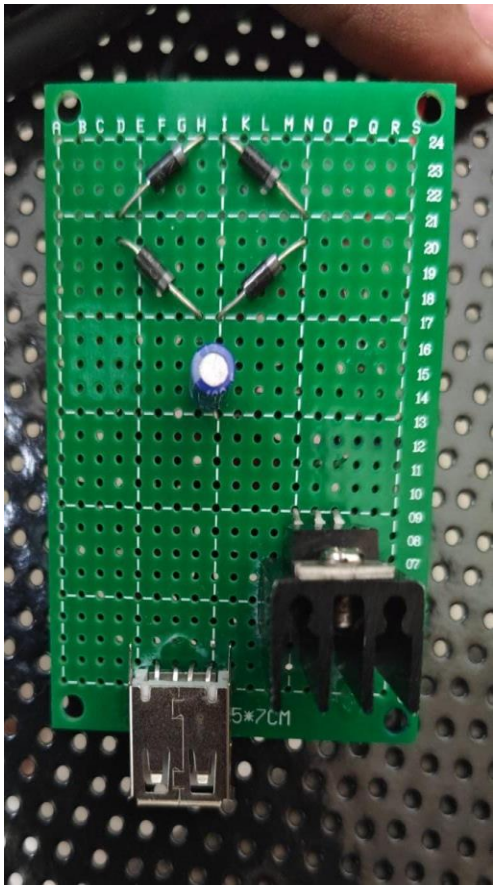
13. Soldering the four pins of the USB socket to the perfboard was challenging because the pins were small and closely spaced, requiring careful handling to ensure secure connections without unwanted solder bridges.

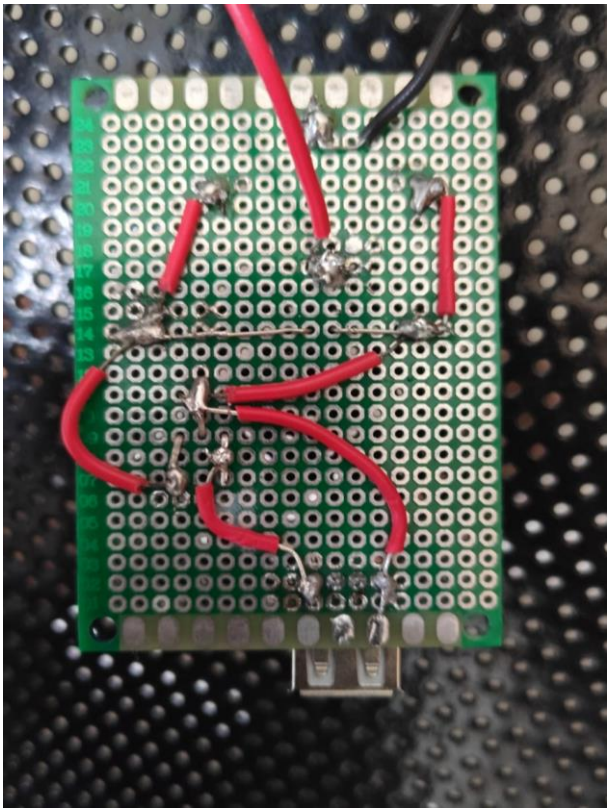
14. The four pins of the USB socket were soldered to the perfboard to provide stability and securely hold the socket in place.

15. The USB socket was not checked for cable compatibility before soldering. After soldering it onto the perfboard, we found that the cable could not be inserted properly because the socket was damaged. Therefore, we desoldered and removed it and replaced it with a new USB socket.

16.Desoldering the damaged USB socket was challenging, but we successfully removed it and replaced it with a new USB socket after checking its compatibility with the cable beforehand.

17.The output of the voltage regulator was connected to the VBUS pin of the USB socket, while the ground (GND) of the voltage regulator was connected to the GND pin of the USB socket.





⚠ Note: 7805 OUT should connect to USB VBUS (+5 V), and 7805 GND should connect to USB GND.

18. The voltage across the VBUS and GND pins of the USB socket was measured using a multimeter to verify that the circuit provided a stable 5 V DC output(approximately).

19. After rechecking all the circuit connections, the USB cable was connected to a mobile phone, and the phone successfully started charging, confirming the proper functioning of the charger circuit.



4. Expected Voltages vs Actual Voltages

Measurement point	Expected values	Actual values
Main input	230 V AC RMS	242 V AC RMS
Transformer secondary	12 V AC RMS	12 V AC RMS
Transformer peak	≈17 V Peak	17 V PEAK
Across smoothing capacitor	≈14-16 V DC	17.2 V DC
7805(input)	Approx the capacitor voltage	17 V DC
7805(output)	≈5 V DC	5 V DC
USB power output	≈5 V DC	5.1 V DC

5. Important note about 7805 :

The 7805 is a linear regulator. If its input is around 15 V and the output is 5 V, the extra voltage is

converted into heat. For example, at 0.5 A the approximate heat is (15−5)×0.5 = 5 W. A heatsink may

therefore be required, and the transformer, diodes, capacitor and regulator must all be rated for the

intended current.

For a real high-current phone charger, a properly designed switch-mode USB power supply/buck

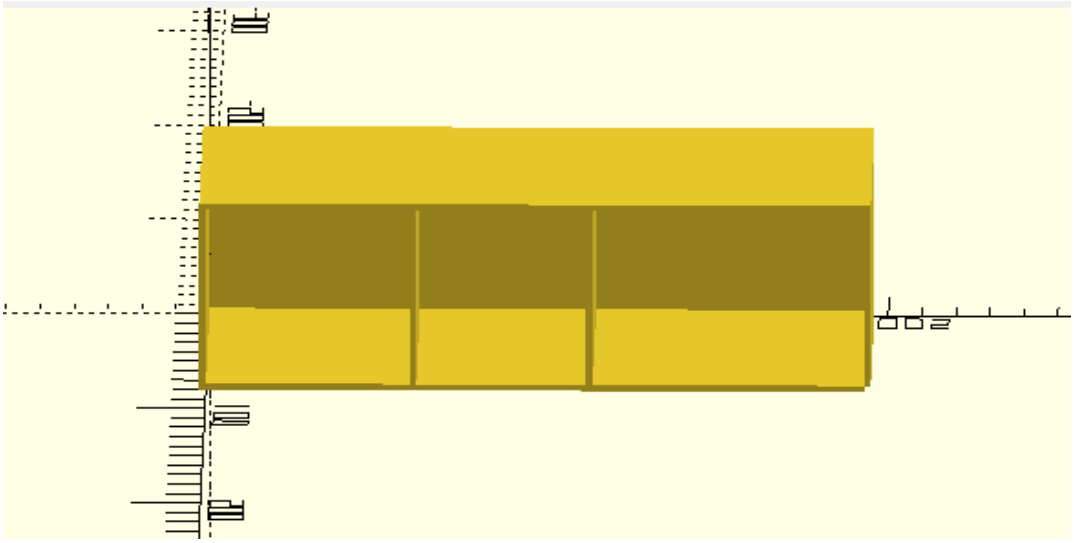
converter is generally more suitable than a 7805-based design.

6. 3D printing-progress log:

6.1

1) Initially, we planned the enclosure as a cuboid with two partitions, dividing it into 3 separate compartments.

2) The first compartment was intended for storing extra cable of transformer, the second for the transformer and the third for the circuit mounted on the perf board.

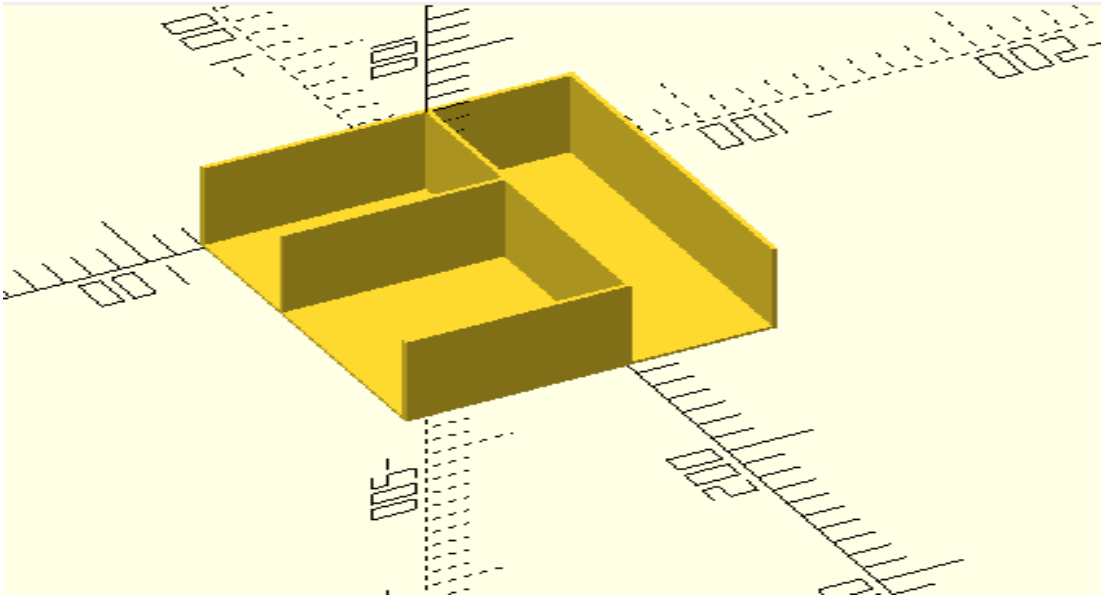


3) As the enclosure was becoming too long, we planned to modify the design to make it compact.

6.2

1) The design was modified by placing the cable compartment beside the two main compartments, increasing the breadth instead of the length. This made the enclosure more compact and provided better space utilization.

2) Routing the transformer cable from the front-left compartment to the back compartment requires cutting holes through the partition and the right back side wall.



3) Sealing and closing the entire enclosure structure at the final stage has become extremely difficult and complex due to the layout and routing requirements.

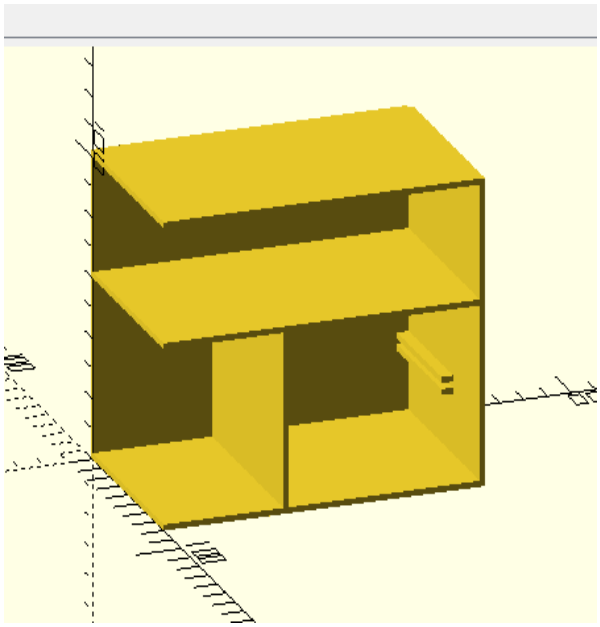
4) So we modified the design again.

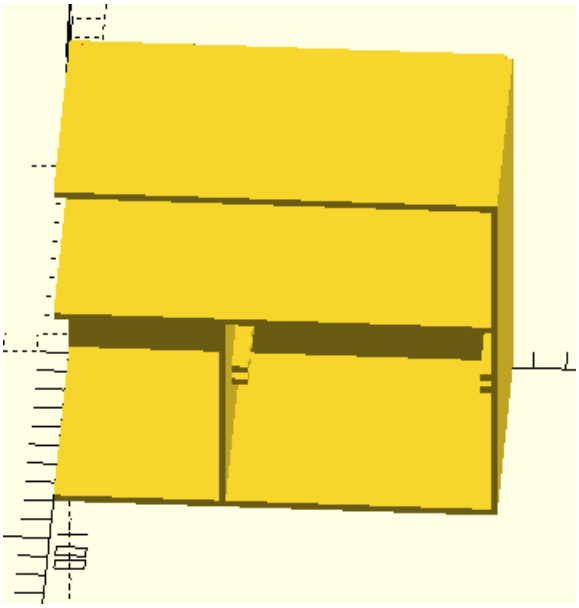
6.3

1) In the next design modification, the additional cable compartment was moved from the side and positioned above the two main compartments.

2) This arrangement was intended to make better use of the available space.

3) We planned to slide the circuit in it's compartment. So we added support/sliding rails in the 3rd compartment.





4) However, the USB socket protruded from the side of the perfboard and became obstructed by the right wall of the enclosure, preventing the perfboard from sliding into its position smoothly as we sliding it in a horizontal way.

6.4 Final Enclosure Design

1) **Vertical Sliding Mechanism:**

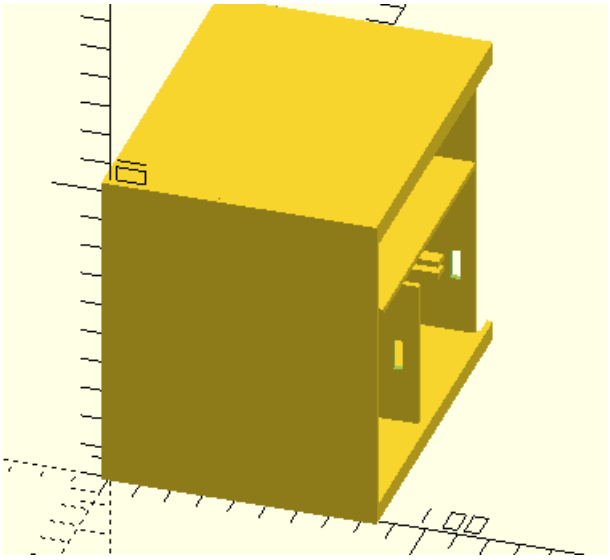
To overcome the sliding obstruction encountered in the horizontal arrangement, the circuit was redesigned to slide into the enclosure vertically instead of horizontally. This provided a smoother insertion mechanism and better utilization of the available space.

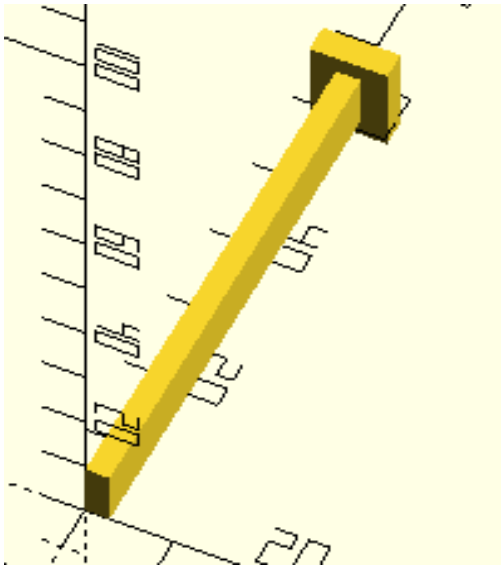
2) **USB Access Opening:**

A dedicated opening was provided on the back wall of the enclosure for the USB socket. This allows the USB connector to extend through the enclosure wall, making it easily accessible for connecting the charging cable.

3) **Circuit Locking Mechanism:**

To prevent the circuit from sliding back out of position, holes were provided in the second and third compartments. A removable locking bar was designed to fit into these holes and securely hold the circuit in place after insertion.





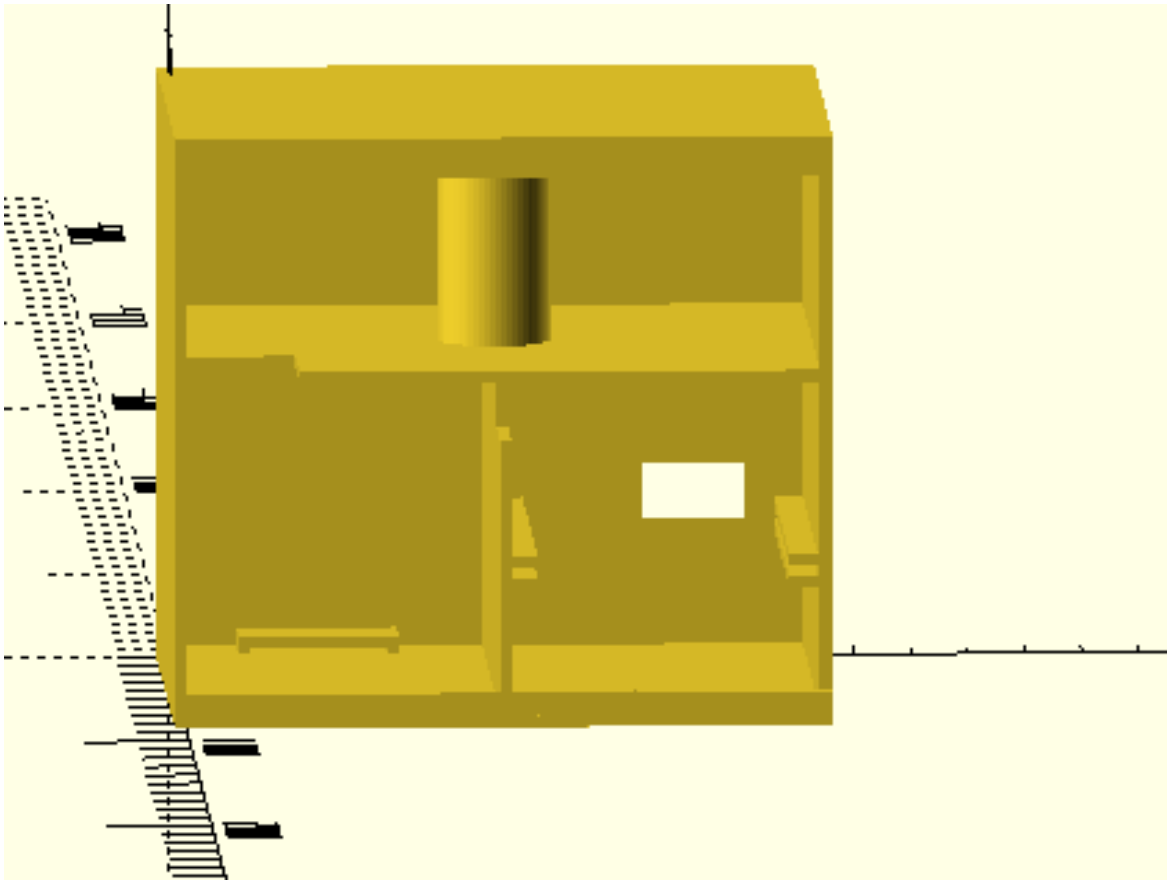
4) A dedicated opening was provided between the first and second compartments at the upper section of the enclosure. This opening allows the wires to pass between the compartments and connect the transformer to the circuit conveniently, while keeping the wiring organized inside the enclosure.

5) A dedicated opening was provided at the top-right corner of the first compartment to allow the transformer wire to pass freely through the enclosure. The wire can be routed upward through this opening .

6) A cylindrical support was placed at the centre of the upper compartment to provide a dedicated space for winding and storing the extra cable. This keeps the cable neatly organized and prevents it from occupying unnecessary space inside the enclosure.

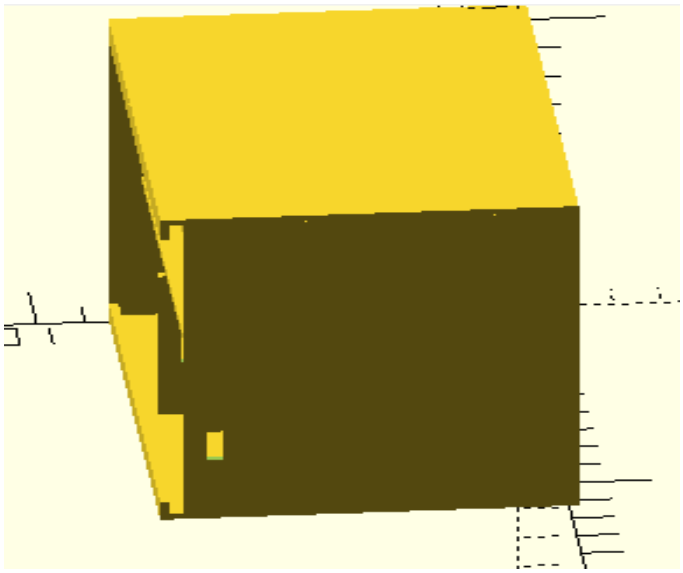
7) An additional support structure was provided in the left compartment to securely hold the transformer in place. This support helps prevent the transformer from shifting during assembly and operation, ensuring that it remains properly positioned inside the enclosure.

8) Initially, we planned to provide two doors, one on the left side and another on the front side, to access the compartments. However, after providing an opening at the interface between the left compartment and the upper compartment, we found that a single front door was sufficient to provide access to the required compartments. Therefore, the left-side door was eliminated, simplifying the overall enclosure design.

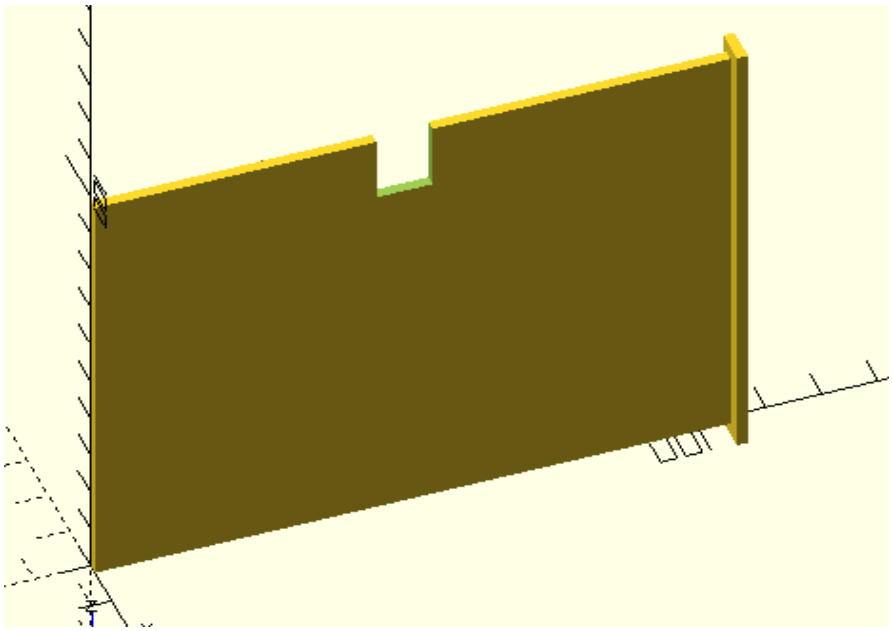


9) Initially, the front door was designed to be fixed to the enclosure using screws. However, incorporating screw fittings made the design and assembly more complex. Therefore, the design was modified by replacing the screw-fixed door with a sliding door mechanism after knowing it is feasible, making the enclosure simpler to assemble and easier to access.

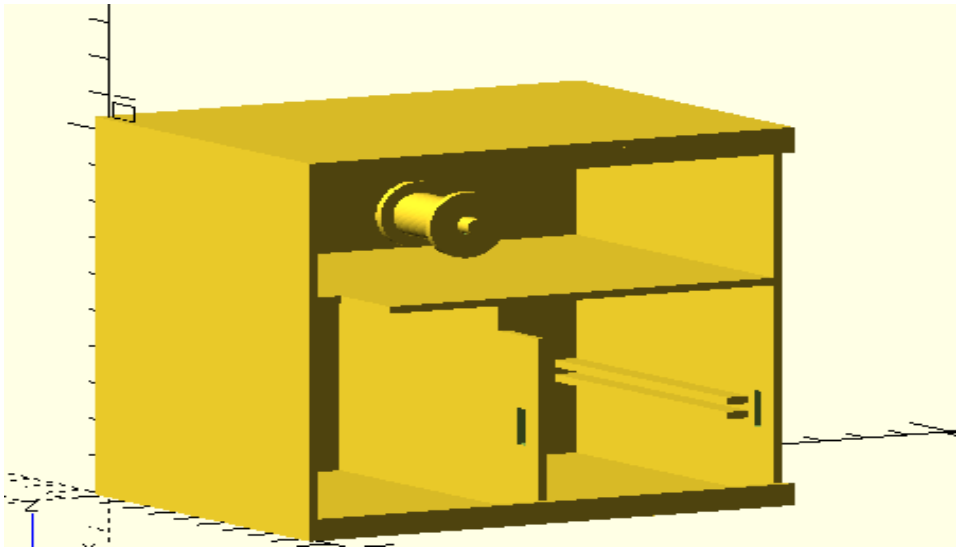
10) An extended guide structure was provided along the sides of the enclosure to create a sliding path for the front door. This allows the door to slide smoothly into position and makes the opening and closing mechanism simple and convenient.



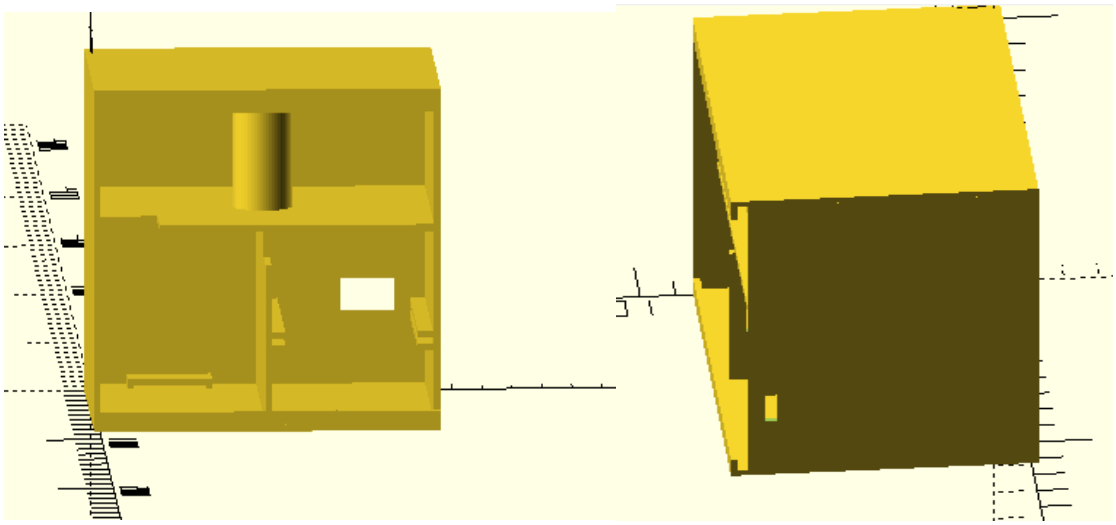
11) A small opening was provided at the top of the sliding door to allow the extra cable to pass through. The cable can be wound around the cylindrical support in the upper compartment and then routed through this opening. This arrangement allows the cable to remain connected while the sliding door is moved into position.



12) We initially planned to use a rotating shaft to roll the wire, but realized it may not be feasible or efficient due to its small size.



13) So, we finally used the cylinder instead of a shaft.



7.References:

a) prof. G. V. V. Sharma (main pdf in github).

b) ChatGPT, Gemini and youtube

c) Circuit connections - Pdf and Chatgpt

Youtube - For learning openscad

Chatgpt – documentation (just for making our sentences with proper vocabulary)